An SQ-Hardness Reduction for Black-Box Attacks on SQ-Implementable Learners

Working note

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Abstract

We give a self-contained reduction showing that, under the planted-clique conjecture, no black-box query-bounded adversary can detect the planted structure of a graph encoded into the training data of an SQ-implementable learner. The reduction is a one-step lift of the Feldman–Grigorescu–Reyzin–Vempala–Xiao 2017 SQ lower bound for planted clique, from *SQ algorithms attacking PC directly* to *black-box adversaries attacking the output of an SQ-implementable learner*. The adversary's query budget does not enter the bound; only the learner's SQ parameters do. Specializations to attribute inference, backdoor activation, membership inference, and model extraction follow as corollaries. Concrete-security calibration and explicit caveats are provided.

1 Setting

1.1 The hard problem (source)

Definition 1 (Planted Clique distribution)**.** For positive integer n and $k \leq n$, the *planted clique distribution* $\mathcal{P}_{n,k}$ is defined as follows. Sample a uniformly random subset $S \subseteq [n]$ with $|S| = k$. Sample a graph G on vertex set $[n]$ where every edge with both endpoints in S is present, and every other edge is independently present with probability $\frac{1}{2}$. The *null distribution* $\mathcal{P}_{n,0}$ is the Erdős–Rényi distribution $G(n, \frac{1}{2})$.

Conjecture 2 (Planted Clique)**.** For any constant $\delta \in (0, \frac{1}{2})$, no polynomial-time algorithm distinguishes $\mathcal{P}_{n,k}$ from $\mathcal{P}_{n,0}$ with constant advantage when $k = n^{1/2-\delta}$.

The Hirahara–Shimizu STOC 2024 equivalence theorem [3] shows that almost all variants of the PC conjecture (search vs. decision, exponentially-close-to-one vs. non-negligible advantage, worst-case k-clique on incompressible graphs) reduce to this conjecture, so we use it without specifying the variant.

1.2 The learning game (target)

Fix n and a clique-size threshold k. Let G be a graph on $[n]$ drawn either from $\mathcal{P}_{n,0}$ or $\mathcal{P}_{n,k}$.

Encoding. Define the *vertex-row encoding* $\mathcal{D}_G$ as the distribution over labeled examples obtained by sampling a vertex v uniformly from $[n]$ and outputting (x_v, y_v) where:

- $x_v = G[v, \cdot] \in \{0, 1\}^n$ is the adjacency row of v;
- y_v is some deterministic function of x_v fixed in advance (e.g., the parity of x_v's coordinates above some index, or simply $y_v = 0$). The choice does not affect the proof so long as y_v is computable from x_v.

$\mathcal{D}_G$ is a distribution over (x, y) pairs whose underlying randomness is the choice of $v \in [n]$. For any function $f\colon \{0,1\}^n \times \{0,1\} \to [0,1]$,

$$\mathbb{E}_{(x,y)\sim\mathcal{D}_G}[f(x,y)] \;=\; \frac{1}{n}\sum_{v\in[n]} f(G[v,\cdot], y_v).$$

This is a *vertex-average* over G — a deterministic function of G alone.

SQ-implementable learner. A learning algorithm L is (q,τ)-*SQ-implementable* if its only access to $\mathcal{D}_G$ consists of at most q queries of the form

> For a given function $f\colon \{0,1\}^n \times \{0,1\} \to [0,1]$, output an estimate $\widehat{a}$ of $\mathbb{E}_{(x,y)\sim\mathcal{D}_G}[f(x,y)]$ within additive tolerance τ.

All other internal computation is data-independent. This is the standard SQ access model of Kearns [5] adapted to a finite-graph-derived distribution. Mini-batch SGD with batch size $B = \Omega(1/\tau^2)$ and ERM with smooth losses are SQ-implementable in this sense. Memorization-prone deep models can also be SQ-implementable as algorithms — implementability is a property of the access pattern, not of the resulting model's expressiveness. Capacity bounds on the resulting model arise as a *consequence* of the SQ access budget ($q \cdot \log(1/\tau)$ bits maximum), not as a separate hypothesis.

The learner L outputs a model $M\colon \mathcal{X} \to \mathcal{Y}$. We assume M is deterministic given the SQ answers; randomized M is handled in Remark 4.

The adversary. A *black-box adversary* A is a randomized algorithm with oracle access to M. A makes at most q_A queries (input choices for which it observes M's output) and outputs a bit $b \in \{0,1\}$. A's advantage in distinguishing $\mathcal{P}_{n,0}$ from $\mathcal{P}_{n,k}$ is

$$\mathrm{adv}(A) \;=\; \Big|\Pr_{G\sim\mathcal{P}_{n,k}}[A^M = 1] \;-\; \Pr_{G\sim\mathcal{P}_{n,0}}[A^M = 1]\Big|,$$

where in each probability $M = L(\mathcal{D}_G)$ is the trained model on the encoded dataset. The probability is over the choice of G, the learner's SQ-answer randomness (within tolerance τ), and A's coins.

2 The Primitive Lower Bound

We use the SQ lower bound for planted clique in its standard form.

Theorem 3 (Feldman–Grigorescu–Reyzin–Vempala–Xiao [1], Theorem 1.1)**.** *Let $\delta > 0$ and $k = n^{1/2-\delta}$. Any SQ algorithm with q queries of tolerance $\tau \ge n^{-1/2+\delta/2}$ distinguishing $\mathcal{P}_{n,k}$ from $\mathcal{P}_{n,0}$ over vertex-SQ access satisfies*

$$q \cdot \tau^2 \;\ge\; \Omega(n^{2\delta}).$$

That is, distinguishing with advantage $\Omega(1)$ requires either $q = \Omega(n^{2\delta})$ queries at constant tolerance, or smaller tolerance $\tau = O(n^{-\delta})$. Equivalently, in the $\mathrm{VSTAT}(t)$ *variant, distinguishing requires $t \ge \Omega(n^{2\delta})$.*

For the secret-leakage variant PC_ρ (where the planted set S is drawn from a leakage distribution ρ rather than uniformly), Brennan–Bresler [2] (Theorem 5.1 and §5.2) establish the same SQ lower bound up to a constant factor.

We encapsulate the bound as a function: define $\mathrm{adv}_{\mathrm{SQ}}(n,k,q,\tau)$ to be the maximum SQ-distinguishing advantage achievable with q queries of tolerance τ. By Theorem 3,

$$\mathrm{adv}_{\mathrm{SQ}}(n,k,q,\tau) \;\le\; O\!\left(\frac{q\cdot\tau^2\cdot n}{k^2}\right),$$

which for $k = n^{1/2-\delta}$ and $\tau = O(1)$ gives

$$\mathrm{adv}_{\mathrm{SQ}}(n,k,q,O(1)) \;\le\; O(q\cdot n^{-1+2\delta}) \;\to\; 0 \quad \text{unless} \quad q=\Omega(n^{1-2\delta}).$$

The exact form of $\mathrm{adv}_{\mathrm{SQ}}$ varies across formulations; what matters for the reduction is that it is the same function on both sides of the inequality below.

3 Main Theorem

Theorem 4 (SQ-Hardness Reduction for Black-Box Attacks)**.** *Let A be a black-box adversary making q_A queries to M. Let L be a deterministic (q_L,τ_L)-SQ-implementable learner with vertex-row encoding $\mathcal{D}_G$. Let G be drawn from $\mathcal{P}_{n,0}$ or $\mathcal{P}_{n,k}$. Then*

$$\mathrm{adv}(A) \;\le\; \mathrm{adv}_{\mathrm{SQ}}(n,k,q_L,\tau_L).$$

In particular, by Theorem 3 for $k = n^{1/2-\delta}$: any SQ-implementable learner with $q_L = \mathrm{poly}(n)$ SQ queries at constant tolerance produces a model M against which no *black-box adversary — regardless of q_A — can achieve constant distinguishing advantage between planted and null.* ***The adversary's query budget does not enter the bound.***

Proof. We construct an SQ algorithm B that distinguishes $\mathcal{P}_{n,0}$ from $\mathcal{P}_{n,k}$ with the same advantage as A but uses only q_L vertex-SQ queries on G with tolerance τ_L. Combined with Theorem 3, this gives the bound.

Step 1: SQ simulation of the learner. Let L's SQ queries be $Q_1, Q_2, \ldots, Q_{q_L}$, where Q_i may depend on the answers of $Q_1,\ldots,Q_{i-1}$ (adaptive SQ). Each query Q_i is a function $f_i\colon \{0,1\}^n \times \{0,1\} \to [0,1]$. By the encoding,

$$\mathbb{E}_{(x,y)\sim\mathcal{D}_G}[f_i(x,y)] \;=\; \frac{1}{n}\sum_{v\in[n]} f_i(G[v,\cdot], y_v),$$

which depends only on the graph G, not on additional randomness in $\mathcal{D}_G$. Thus a vertex-SQ query on G with the same function f_i and tolerance τ_L returns an answer that is, by construction, identically distributed to the SQ answer L would have received from $\mathcal{D}_G$.

Step 2: B's construction. B is given vertex-SQ access to G. B simulates L:

(1) For $i = 1$ to q_L, B issues vertex-SQ query Q_i (computed adaptively from previous answers), receives a_i within tolerance τ_L of the true value.

(2) After all queries, B reconstructs $\widehat{M} = L(a_1,\ldots,a_{q_L})$ — the model L would have output given the same SQ answers. Since L is deterministic given its SQ answers, $\widehat{M} = M$ (the actual model L would produce on $\mathcal{D}_G$ with the same SQ randomness).

(3) B then simulates A: B has full access to $\widehat{M}$ and can answer A's q_A oracle queries by directly evaluating $\widehat{M}$ on A's chosen inputs, using *no further SQ queries on G*. B outputs whatever A outputs.

Step 3: Advantage equivalence. B's output, conditioned on SQ-answer randomness within tolerance τ_L, equals A's output (when A is run on $\widehat{M} = M$). Marginalizing over both algorithms' coins,

$$\mathrm{adv}(B) \;=\; \Big| \Pr_{G\sim\mathcal{P}_{n,k}}[B=1] \;-\; \Pr_{G\sim\mathcal{P}_{n,0}}[B=1] \Big| \;=\; \mathrm{adv}(A).$$

Step 4: Apply Theorem 3. B is an SQ algorithm with q_L vertex-SQ queries of tolerance τ_L distinguishing $\mathcal{P}_{n,0}$ from $\mathcal{P}_{n,k}$. By Theorem 3,

$$\mathrm{adv}(B) \;\leq\; \mathrm{adv}_{\mathrm{SQ}}(n,k,q_L,\tau_L).$$

Combining with $\mathrm{adv}(A) = \mathrm{adv}(B)$ gives the claim. □

4 Remarks

Remark (The adversary's query budget is decoupled from the bound). The bound depends on q_L and τ_L (learner's parameters), not on q_A. A can make exponentially many queries to M and still cannot exceed the SQ-PC distinguishing limit fixed by L's data access. The information channel from G to M is what bounds the attacker's advantage — not the attacker's compute. The mathematical reason is that A's queries to M are computable from M itself (post-training); they do not access G further.

This is the mechanism underlying the "M is determined by its SQ summary" argument and is the core formal reason why SQ-implementability plus a bounded learner-query-budget gives security against any post-training attack in the black-box model.

Remark (Capacity bound is implicit). A common framing introduces "bounded model capacity C" as a separate hypothesis. In this proof, $C \leq q_L \cdot \log(1/\tau_L)$ bits is automatic — M is reconstructed from q_L SQ answers, each providing $\log(1/\tau_L)$ bits beyond the prior. No separate capacity hypothesis is needed.

Remark (Specializations to standard attack classes). The theorem gives a primitive distinguishing-advantage bound. Specific attacks correspond to specific decision rules for A.

- *Attribute inference.* A bit $\alpha(G)$ of G's structure (e.g., "is vertex 1 in the planted clique?") is the target. A guesses $\alpha(G)$ from M-queries. By the theorem with H_1 conditioned on $\alpha = 1$ and H_0 conditioned on $\alpha = 0$, A's advantage over random guessing is bounded by $\mathrm{adv}_{\mathrm{SQ}}(n,k,q_L,\tau_L)$.
- *Backdoor activation.* A trigger input x^* activates pre-planted behavior in M iff G has the planted clique. A finds x^* by querying M. Equivalent to attribute inference with $\alpha = \mathbf{1}[G \text{ has planted clique}]$, so the same bound applies.
- *Membership inference.* "Is point (x_v, y_v) in the training set?" Encoded as an attribute α of G via the vertex v's adjacency row, this reduces to attribute inference.
- *Model extraction.* A reconstructs an approximation $\widehat{M}_A$ of M via queries. Information about G in $\widehat{M}_A$ is bounded by information about G in M, which is bounded by $q_L \cdot \log(1/\tau_L)$ bits. Specifically,

$$I(G;\widehat{M}_A) \;\leq\; I(G;M) \;\leq\; q_L \cdot H(\text{SQ answer} \mid \text{query}) \;\leq\; q_L \cdot \log(1/\tau_L),$$

  so any G-property A extracts from $\widehat{M}_A$ inherits the same SQ-PC bound.

Remark (Randomized learners). If L uses internal randomness r in addition to its SQ answers, the simulation in the proof must condition on r. Either (a) r is public (visible to all parties; in particular B samples r from the same distribution and proceeds), or (b) r is private. In case (b), L's output M is a function of $(a_1, \ldots, a_{q_L}, r)$. B samples r from the same distribution and the rest of the proof proceeds with one extra averaging step over r. The bound is unchanged.

Remark (Where the proof fails). The proof breaks if any of the following conditions are violated:

- **L is not SQ-implementable.** Memorization through individual-record access (kNN, hash tables, retrieval-augmented LMs that store-and-retrieve verbatim training records) is the most common way out. A non-SQ-implementable learner can encode information about specific G-edges into M that is not bounded by $\mathrm{adv}_{\mathrm{SQ}}$.
- **The encoding is not vertex-row-shaped.** Encodings using, e.g., random projections of adjacency rows may dilute G's structure into the SQ summary in a way that strengthens or weakens the bound — a separate analysis.
- **White-box adversary** (sees M's weights). The proof does not bound non-SQ functionals of M's weights. White-box attacks can in principle extract more, and a separate cryptographic assumption (e.g., indistinguishability obfuscation [6]) is needed to rule them out.

Remark (Concrete-security calibration). The asymptotic bound becomes concrete by plugging in. For $n = 10^4$ and $\delta = 0.1$ (so $k = n^{0.4} \approx 40$), Theorem 3 says distinguishing requires $q \cdot \tau^2 \geq \Omega(n^{0.2}) \approx 6.3$. A learner with $q_L = 10^5$ and $\tau_L = 10^{-3}$ has $q_L \cdot \tau_L^2 = 0.1$ — well below the threshold — so $\mathrm{adv}(A)$ is bounded by $O(0.1/6.3) \approx 0.016$. The adversary's distinguishing advantage is at most 1.6%.

This is a working-point security claim, parameterizable: smaller k, larger n, smaller q_L, larger τ_L all tighten the bound. Practical learners ($q_L \approx 10^5$ to 10^7, $\tau_L \approx 10^{-3}$ to 10^{-5}) at modest n give meaningful bounds.

5 Caveats

This proof does *not* establish the following.

- **It is not a hardness proof for any specific deployed attack.** It is a hardness proof for *information-theoretic distinguishing* under the SQ-implementable plus black-box assumption. Real-world attacks may rely on non-SQ-implementable learners or white-box access; these need separate treatment.
- **It does not give a tight bound** in the regime where q_L is very small or τ_L very large. The bound is non-trivial when $q_L \cdot \tau_L^2 < n^{2\delta}$; outside that regime the SQ-PC bound is vacuous.
- **It does not transfer to LWE-encoded constructions.** The vertex-row encoding is tied to PC's combinatorial structure; constructions using lattice-based encodings need a different reduction (and the entropy mismatch with learned weights re-enters).
- **It does not handle the per-round federated-learning observation model.** That requires composition over T rounds and a different machinery — most likely Mahloujifar's generalized p-tampering [7] adapted for PC-encoded learning.

References

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